Li Hu

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EDUCATION

University of Science & Technology of China Sep. 2024 – Jun. 2027
School of Cyber Science and Technology | Master of Science in Cybersecurity Hefei, China

- GPA: 3.79/4.30, TOEFL: 4.5/6.0
- Supervisor: Prof. Weiming Zhang, Assoc.Prof. Shaoyin Cheng

Hainan University Sep. 2020 – Jun. 2024
School of Cyberspace Security | Bachelor of Science in Information Security Haikou, Hainan

- GPA: 3.82/4.00, Rank: 2/114

RESEARCH PUBLICATIONS

First-Author and Corresponding-Author Publications (*Co-1st / †Corresponding)

- **[ACL'25]** **Li Hu***, Guoqiang Chen*, Xiuwei Shang, Shaoyin Cheng, Benlong Wu, Gangyang Li, Xu Zhu, Weiming Zhang and Nenghai Yu, "CompileAgent: Automated Real-World Repo-Level Compilation with Tool-Integrated LLM-based Agent System." *The 63rd Annual Meeting of the Association for Computational Linguistics, Main Track, Vienna, Austria* **(CCF-A)**
- **[ASE'26]** Ruijie Zhang*, Huaxiao Yin*, **Li Hu*†**, Fanqi Kong, Haozhe Liang, Shaoyin Cheng, Kun Zhang, and Bibo Tu, "RegoMender: A Multi-Agent Framework for Automated Rego Policy Repair." *The 41st IEEE/ACM International Conference on Automated Software Engineering* **(CCF-A)**
- **[TSE'26]** **Li Hu**, Xiuwei Shang, Jieke Shi, Shaoyin Cheng, Junqi Zhang, Gangyang Li, Zhou Yang, Weiming Zhang, and David Lo, "What Makes LLMs Good Binary Deobfuscators?" *IEEE Transactions on Software Engineering* **(Major Revision, CCF-A)**
- **[TDSC'26]** **Li Hu***, Yanming Zhang*, Xiuwei Shang, Shaoyin Cheng, Junqi Zhang, Guoqiang Chen, Weiming Zhang, and Nenghai Yu, "Digging in the Shadows: Detecting Covert-Communication Functions in Software Binaries via Decoder-Only LLMs." *The IEEE Transactions on Dependable and Secure Computing* **(Under Review, CCF-A)**
- **[ICSE'27]** **Li Hu**, Xiuwei Shang, Gangyang Li, Xu Zhu, Shaoyin Cheng, Junqi Zhang, Guoqiang Chen, Weiming Zhang, and Nenghai Yu, "Lifting the Veil: Reasoning-Enhanced LLMs for Binary Deobfuscation via Reinforcement Learning." *The 49th International Conference on Software Engineering* **(Under Review, CCF-A)**

Co-authored Publications

- **[ASE'26]** Xiuwei Shang, **Li Hu**, Xiao Jiang, Jieke Shi, Junda He, Zhou Yang, Shaoyin Cheng, Guoqiang Chen, Weiming Zhang, and David Lo, "Beyond Text Matching: Towards Reference-Free Evaluation for Human-Oriented Binary Reverse Engineering." *The 41st IEEE/ACM International Conference on Automated Software Engineering* **(CCF-A)**
- **[EMSE'26]** Xiuwei Shang, **Li Hu**, Shaoyin Cheng, Shikai Guo, Guoqiang Chen, Benlong Wu, Weiming Zhang, and Nenghai Yu, "Binary Code Similarity Detection via Graph Contrastive Learning on Intermediate Representations." *Empirical Software Engineering* **(Major Revision, CCF-B)**
- **[TSE'27]** Zhenkan Fu, Xiuwei Shang, **Li Hu**, Shaoyin Cheng, Gangyang Li, Zhengdao Li, Shikai Guo, Weiming Zhang, and Nenghai Yu, "Knowledge-Enhanced In-Context Learning for Multi-Intent Binary Code Summarization." *The IEEE Transactions on Software Engineering* **(Under Review, CCF-A)**
- **[TIFS'27]** Xu Zhu, Shaoyin Cheng, Junqi Zhang, **Li Hu**, Gangyang Li, Weiming Zhang, and Nenghai Yu, "Danger Around the Corner: Unveiling Cross-Coupling Vulnerabilities in EKF Localization During Turns." *IEEE Transactions on Information Forensics and Security* **(Under Review, CCF-A)**
- **[NeurIPS'26]** Ruijie Zhang, Haozhe Liang, Da Chang, **Li Hu**, Fanqi Kong, Huaxiao Yin, and Yu Li, "When Does Value-Aware KV Eviction Help? A Fixed-Contract Diagnostic for Non-Monotone Cache Compression." *The 40th Annual Conference on Neural Information Processing Systems* **(Under Review, CCF-A)**
- **[CCS'26]** Gangyang Li, Xiuwei Shang, Xu Zhu, **Li Hu**, Shaoyin Cheng, Junqi Zhang, Weiming Zhang, and Nenghai Yu, "ReStruct: Three-Stage Binary Data Structure Recovery via Inter-procedural Analysis and RL-Enhanced LLMs." *The 33rd ACM Conference on Computer and Communications Security* **(Under Review, CCF-A)**

- **[ASE'25]** Gangyang Li, Xiuwei Shang, Shaoyin Cheng, Junqi Zhang, **Li Hu**, Xu Zhu, Weiming Zhang, and Nenghai Yu, "PseudoFix: Refactoring Distorted Structures in Decompiled C Pseudocode." *The 40th IEEE/ACM International Conference on Automated Software Engineering* **(CCF-A)**
- **[TOSEM'26]** Gangyang Li, Xiuwei Shang, Shaoyin Cheng, Junqi Zhang, **Li Hu**, Xu Zhu, Weiming Zhang, and Nenghai Yu, "Beyond the Edge of Function: Unraveling the Patterns of Type Recovery in Binary Code." *ACM Transactions on Software Engineering and Methodology* **(CCF-A)**
- **[IJCAI'25]** Xiuwei Shang, Guoqiang Chen, Shaoyin Cheng, Benlong Wu, **Li Hu**, Gangyang Li, Weiming Zhang, and Nenghai Yu, "BinMetric: A Comprehensive Binary Code Analysis Benchmark for Large Language Models." *The 34th International Joint Conference on Artificial Intelligence* **(CCF-A)**
- **[ICSME'24]** Xiuwei Shang, Shaoyin Cheng, Guoqiang Chen, Yanming Zhang, **Li Hu**, Xiao Yu, Weiming Zhang, and Nenghai Yu, "How Far Have We Gone in Binary Code Understanding Using Large Language Models." *The 40th International Conference on Software Maintenance and Evolution* **(CCF-B)**
- **[NeurIPS'26]** Guoqiang Chen, Lingyun Ying, Ziyang Song, Daguang Liu, Qiang Wang, **Li Hu**, Shaoyin Cheng, Weiming Zhang, and Nenghai Yu, "Cross-modal Retrieval Models for Stripped Binary Analysis." *The 40th Annual Conference on Neural Information Processing Systems* **(Under Review, CCF-A)**
- **[EMSE'25]** Xiuwei Shang, Zhankan Fu, Shaoyin Cheng, Guoqiang Chen, Gangyang Li, **Li Hu**, Weiming Zhang, and Nenghai Yu, "An Empirical Study on the Effectiveness of Large Language Models for Binary Code Understanding." *Empirical Software Engineering* **(CCF-B)**

ACADEMIC SERVICES

Reviewer

- Engineering Applications of Artificial Intelligence (EAAI), 2026
- ACM Transaction on Software Engineering and Methodology (TOSEM), 2025
- ACL Rolling Review (ARR), 2025

HONORS & AWARDS

- **China National Scholarship**, University of Science & Technology of China, 2026
- **First-class Academic Scholarship**, University of Science & Technology of China, 2026
- **First-class Academic Scholarship**, University of Science & Technology of China, 2025
- **Information Security and Countermeasures Contest**, National Second Prize, 2025
- **First-class Academic Scholarship**, University of Science & Technology of China, 2024
- **DataCon Vulnerability Analysis Track**, Third Prize, 2024
- **Outstanding Graduates of Hainan University**, Hainan University, 2024
- **Excellent Undergraduate Thesis (Top 2%)**, Hainan University, 2024
- **National Undergraduate Mathematics Competition**, National Second Prize, 2023
- **Chang'an Cup Electronic Data Forensics Competition**, National Third Prize, 2023
- **China Undergraduate Mathematical Contest in Modeling (CUMCM)**, Province Second Prize, 2023

VULNERABILITY DISCOVERIES

- **Selected High-Impact Vulnerabilities:**
  - **CVE-2025-13400 [CVSS 9.8 Critical]**: Discovered a buffer overflow vulnerability in *Tenda CH22 1.0.0.1*, affecting *formWrlExtraGet* in */goform/WrlExtraGet* via the *chkHz* parameter.
  - **CVE-2025-11423 [CVSS 9.8 Critical]**: Discovered a memory corruption vulnerability in *Tenda CH22 1.0.0.1*, affecting *formSafeEmailFilter* in */goform/SafeEmailFilter* via the *page* parameter.
- **Other Confirmed Vulnerabilities (10+ Discoveries):**
  - **Representative CVE Numbers:** CVE-2025-11505, CVE-2025-11507, CVE-2025-10079
  - **Representative CNVD Numbers:** CNVD-2025-27358, CNVD-2026-13451, CNVD-2026-14713